Solution

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
Second Semester, Session: 2025 – 2026
Control Systems (ECE/EEE/INSTR F242)

MM: 06 **Time: 12 Minutes** **Quiz - 3** **Date 24.02.2026**

Note: Write your final answers in the space provided only. Show all the necessary steps in solution.

ID: Solution **Name:** Solution **Section:** Solution.

1. Consider a transfer function $G(s) = 2/(s+5)$, find the time taken by this system to reach 98% of its steady state value, if a unit step input is applied to this system. **[2 M]**

$G(s) = \frac{2}{(s+5)}$ $g = \frac{0.4}{(1+0.2s)}$ $\Rightarrow c(t) = 0.4(1-e^{-t/0.2})$ $(\because T = 0.2 sec)$,

for 98%, $0.98 \times 0.4 = 0.4(1-e^{-t_0/0.2})$ (1M)

$\Rightarrow t_0 = 0.7824 sec \approx 0.8 sec$

Time taken by system to reach 98% of its final value = 0.7824 or 0.8 sec (4T) — (1M)

2. A spring-mass-damper system has the following transfer function,

$$\frac{X(s)}{F(s)} = \frac{1}{(s^2+10s+100)}$$

Find, ***(i)*** the value of damping ratio (ζ), and ***(ii)*** the peak value of displacement $X(t)$, if an input force, $F(t)$ = 20 u(t) N is applied to this system. **[2 M]**

$\frac{X(s)}{F(s)} = \frac{1}{100}\left[\frac{100}{s^2+10s+100}\right]$; $\omega_n = 10$; $2\xi\omega_n = 10 \Rightarrow \xi = 0.5$.

$M_p(\%) = \left[e^{-\xi\pi/\sqrt{1-\xi^2}}\right] \times 100 = 0.163 \times 100 = 16.3\%$

also, $c(\infty) = \frac{20}{100} = 0.2$; $c(t_p) = 1.163 \times 0.2$
$= 0.2326$.

(*i*) Damping ratio (ζ) = 0.5 — (1M)

(*ii*) peak value of displacement (in m) = 0.2326. — (1M)

3. Pole pair (s_1 and s_2) of a second order system are shown in Fig. Q3. By replacing the pole locations, one can reduce the system's peak time (t_p), without altering its settling time. Which of the new pole locations would you recommend from the given choices: (s_{1A}, s_{2A}), encircled by 'Δ', **or** (s_{1B}, s_{2B}), encircled by '$\square$'? Briefly justify your answer with proper reason. No marks without proper justification. **[2 M]**

Tick the correct choice: s_{1A}, s_{2A} ✓ | s_{1B}, s_{2B} ☐

Justification: $\because$ $t_p = \pi/\omega_d$.

for pole pairs s_{1A}, s_{2A}; the value of $\xi\omega_n$ remain same & ω_d (or ω_n) increases, so, peak time will reduce & settling time will ~~incre~~ remain same.

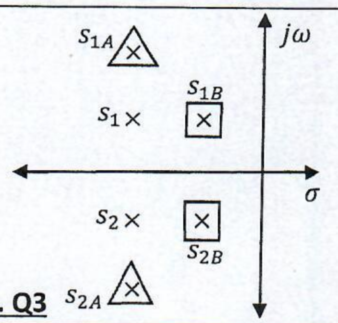

Fig. Q3

2m — only if reason is also correct.